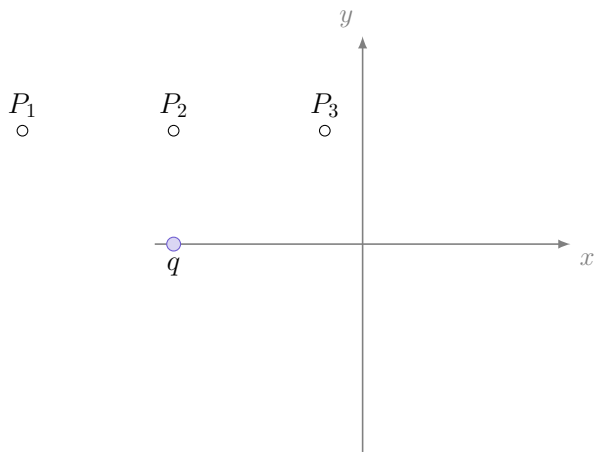
Intro to Biot–Savart

Goal.

- Warm up with something you already know: superpose the $\vec{E}$ field of a point charge at three locations, being explicit about separation vectors and unit vectors.
- Reuse that exact same geometry to introduce the Biot–Savart law for a single *moving* point charge, $\vec{B}$.
- Compare $\vec{E}$ and $\vec{B}$ at those same three points — same source, same locations, very different field.
- Scale up to three moving charges in a line and see their $\vec{B}$ contributions add (not cancel) at an off-axis point.
- Make the leap from one moving charge ($q\vec{v}$) to a continuous current ($I\,d\vec{\ell}$) — the form of Biot–Savart we'll actually use for wires, loops and solenoids.

1. Warm-up: $\vec{E}$ from a point charge

A charge $q = +2.0\ \mu\text{C}$ sits at $(-5, 0, 0)$ cm. Find $\vec{E}$ at three field points: $P_1 = (-9, 3, 0)$, $P_2 = (-5, 3, 0)$, $P_3 = (-1, 3, 0)$ (all in cm).



Sketch the three separation vectors $\vec{r}_1, \vec{r}_2, \vec{r}_3$ from q to each field point on the diagram above.

Step 1 — separation vectors $\vec{r}_i = \vec{r}_{P_i} - \vec{r}_q$ (cm):

$\vec{r}_1 = (_____,\ _____,\ _____)$

$\vec{r}_2 = (_____,\ _____,\ _____)$

$\vec{r}_3 = (_____,\ _____,\ _____)$

Step 2 — magnitudes and unit vectors $\hat{r}_i = \vec{r}_i / r_i$:

$r_1 = _____$ cm, $\hat{r}_1 = (_____,\ _____,\ _____)$

$r_2 = _____$ cm, $\hat{r}_2 = (_____,\ _____,\ _____)$

$r_3 = _____$ cm, $\hat{r}_3 = (_____,\ _____,\ _____)$

Step 3 — Coulomb's law, $\vec{E} = \dfrac{1}{4\pi\varepsilon_0}\dfrac{q}{r^2}\hat{r}$:

$\vec{E}_1 = (_____,\ _____,\ _____)$ N/C

$\vec{E}_2 = (_____,\ _____,\ _____)$ N/C

$\vec{E}_3 = (_____,\ _____,\ _____)$ N/C

Check. P_1 and P_3 are the same distance from q. What's the same about $\vec{E}_1$ and $\vec{E}_3$, and what's different? Why?

2. Biot–Savart: the same charge, now moving

Same charge, same three points — but now q moves with velocity $\vec{v} = (12, 0, 0)$ m/s. The Biot–Savart law for a single point charge is

$$\vec{B} = \frac{\mu_0}{4\pi}\frac{q\,\vec{v}\times\hat{r}}{r^2}, \qquad \frac{\mu_0}{4\pi} = 1\times10^{-7}\ \text{T·m/A}.$$

Reuse your $\hat{r}_1, \hat{r}_2, \hat{r}_3$ from §1 — the geometry hasn't changed, only the source. Compute $\vec{v}\times\hat{r}_i$ first, then $\vec{B}_i$:

$\vec{v}\times\hat{r}_1 = (_____,\ _____,\ _____)$ m/s

$\vec{v}\times\hat{r}_2 = (_____,\ _____,\ _____)$ m/s

$\vec{v}\times\hat{r}_3 = (_____,\ _____,\ _____)$ m/s

$\vec{B}_1 = (_____,\ _____,\ _____)$ T

$\vec{B}_2 = (_____,\ _____,\ _____)$ T

$\vec{B}_3 = (_____,\ _____,\ _____)$ T

Check. Is $\vec{B}_1$ the mirror image of $\vec{B}_3$, the way $\vec{E}_1$ and $\vec{E}_3$ were? What's actually true about $\vec{B}_1$ and $\vec{B}_3$?

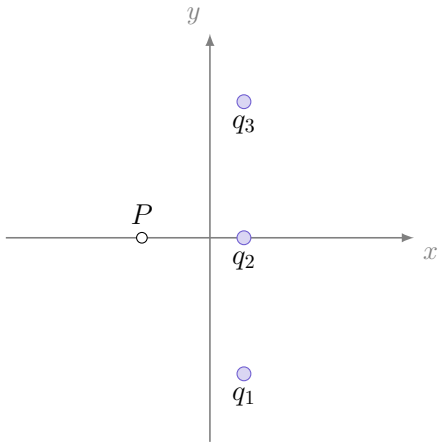
3. Compare $\vec{E}$ and $\vec{B}$

- Which field is bigger here, $\vec{E}$ or $\vec{B}$ — by roughly how many orders of magnitude? _____
- $\vec{E}_1, \vec{E}_2, \vec{E}_3$ point in _____ directions (how do they relate to $\hat{r}$?).
- $\vec{B}_1, \vec{B}_2, \vec{B}_3$ all point _____ (what direction, and why does every one of them end up that way)?

Check. $\vec{v}$ points along $+\hat{x}$ and every field point here has $z = 0$. Could $\vec{B}$ ever have come out with a nonzero x- or y-component at these points? Why or why not?

4. Three moving charges in a line

Three charges, each $q = +1.0\ \mu\text{C}$, sit at $(1, -4, 0)$, $(1, 0, 0)$, $(1, 4, 0)$ cm, each moving with $\vec{v} = (0, 10, 0)$ m/s. Find the total $\vec{B}$ at field point $P = (-2, 0, 0)$ cm — off to the side, not on the line of charges.



Sketch the three separation vectors from each charge to P.

$\vec{r}_1 = (____, ____, ____)$ $r_1 = ____$ cm $\hat{r}_1 = (____, ____, ____)$

$\vec{r}_2 = (____, ____, ____)$ $r_2 = ____$ cm $\hat{r}_2 = (____, ____, ____)$

$\vec{r}_3 = (____, ____, ____)$ $r_3 = ____$ cm $\hat{r}_3 = (____, ____, ____)$

Each charge's contribution $\vec{B}_i = \dfrac{\mu_0}{4\pi}\dfrac{q\,\vec{v}\times\hat{r}_i}{r_i^2}$:

$\vec{B}_1 = (____, ____, ____)$ T $\quad$ $\vec{B}_2 = (____, ____, ____)$ T

$\vec{B}_3 = (____, ____, ____)$ T

$\vec{B}_{\text{total}} = \vec{B}_1 + \vec{B}_2 + \vec{B}_3 =$

Check. Did the three contributions partly cancel, or did they add? Compare to what you'd expect from three point *charges'* $\vec{E}$ fields at an off-axis point — is $\vec{B}$ behaving the same way?

5. From one moving charge to a current

A current is just a huge number of moving charges, confined to a wire. Take a short piece of wire of length $d\ell$, carrying current I: in time dt, the charge that passes through it is dq, moving at the drift velocity $\vec{v}$ along the wire. The replacement rule is

$$q\,\vec{v} \longrightarrow _____$$

so the Biot–Savart law for one point charge, $\vec{B} = \dfrac{\mu_0}{4\pi}\dfrac{q\,\vec{v}\times\hat{r}}{r^2}$, becomes the law for one current element:

$d\vec{B} =$

Check. Everything you did in §1–§4 above with separation vectors and unit vectors — does any of it change once $q\vec{v}$ becomes $I\,d\vec{\ell}$? What's actually new?